

CS & IT ENGINEERING



Digital Logic

Miscellaneous Topics

Lecture No. 2



By- Aditya sir

Recap of Previous Lecture



Topic



Number System

$$(\quad)_b \rightarrow (?)_{10}$$

$$(\quad)_{10} \rightarrow (\quad)_b$$

$$(\quad)_{b_1} \rightarrow (\quad)_{b_2}$$

Topics to be Covered



Topic



Unsigned

Signed

→ 1's Complement Rep_r

→ 2's Complement Rep_r



About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Number System



Binary Coded
Octal
 $2^3 \uparrow$ (3 bits)

Binary
Code Hexad
(4 bits) 2^4

$\uparrow$ 4 bits $\rightarrow +3$

Decimal Digits	BCD 8421	Excess - 3	Octal digits	<u>BCO</u>	Hexadecimal Digits	BCH
0	0000	0011	0	000	0	0000
1	0001	0100	1	001	1	0001
2	0010	0101	2	010	2	0010
3	0011	0110	3	011	3	0011
4	0100	0111	4	100	4	0100
5	0101	1000	5	101	5	0101
6	0110	1001	6	110	6	0110
7	0111	1010	7	111	7	0111
8	1000	1011			8	1000
9	1001	1100			9	1001



Topic : Number System

Decimal Digits	BCD 8421	Excess - 3	Octal digits	BCO	Hexadecimal Digits	BCH
					A	1010
					B	1011
					C	1100
					D	1101
					E	1110
					F	1111



Topic : Number System

x
 $(x-1)$'s Comp
 x 's Comp

Magnitude Representation

Unsigned

Signed

Complement

Representation

only for
pos nos
(+5)

-5

101

(NA)

MSB $\rightarrow$ Sign
1 $\rightarrow$ -ve, 0 $\rightarrow$ +ve
0101

1101

1's complement
(as it is)

0101

1010

2's complement

(as it is)

0101

1011



Topic : Types of Counter

$$r = 10$$

$$r - 1$$

$$r$$

9's complement

10's complement

9's complement
of 537

$$\begin{array}{r} 999 \\ - 537 \\ \hline 462 \end{array}$$



1's Complement of 5?

Flip

0101 $\Rightarrow +5$

1010 $\Rightarrow -5$

$$\begin{array}{r} 1111 \\ - 0101 \\ \hline 1010 \\ \hline \end{array}$$



Topic : Unsigned Representation

3 Bit number:

Range : $0 \rightarrow 7$

: 0 to $2^n - 1$

Range

Unsigned

Binary	Decimal
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7



Topic : Signed Representation

4 Bit number:

Range : -7 → +7

Signed

n Bit number:

Range : $-\{2^{n-1} - 1\}$ to $\{2^{n-1} - 1\}$

Signed Number	Equivalence Decimal
0000	+0
0001	+1
0010	+2
0011	+3
0100	+4
0101	+5
0110	+6
<u>0</u> 111	+7

Signed Number	Equivalence Decimal
<u>1</u> 000	-0
1001	-1
1010	-2
1011	-3
1100	-4
1101	-5
1110	-6
<u>1</u> 111	-7



Topic : 1's Complement Representation

4 Bit number:

Range : $-7 \rightarrow +7$

8 $\rightarrow$ 01000

n Bit number:

Range : $-\{2^{n-1} - 1\}$ to $\{2^{n-1} - 1\}$

1's Comp.	Equivalence Decimal
<u>0</u> 000 $\rightarrow$	+0
<u>0</u> 001 $\rightarrow$	+1
<u>0</u> 010	+2
<u>0</u> 011	+3
0100	+4
0101	+5
0110	+6
0111 $\rightarrow$	+7

1's Comp.	Equivalence Decimal
<u>1</u> 000	-7
<u>1</u> 001 $\rightarrow$ 0110	-6
<u>1</u> 010 $\rightarrow$ 0101	-5
1011 $\rightarrow$ 0100	-4
1100 $\rightarrow$ 0011	-3
1101 $\rightarrow$ 0010	-2
1110 $\rightarrow$ 0001	-1
1111 $\rightarrow$ 0000	-0

0111 $\rightarrow +7$

-7

0111



1000

1000

0111

-7

[NAT]



#Q. $001101 \rightarrow +13$

[NAT]



#Q. 110010 → -13 ✓

~~001101~~ → +13 ✓

1's Comp Rep_r



Topic : 2's Complement Representation

4 Bit number:

Range : -8 → +7

2's Comp.	Equivalence Decimal
<u>0000</u> →	<u>+0</u>
<u>0001</u> →	+1
<u>0010</u> →	+2
0011	+3
0100	+4
0101	+5
0110	+6
0111 →	+7

n Bit number:

Range : $-\{2^{n-1}\}$ to $\{2^{n-1} - 1\}$

2's Comp.	Equivalence Decimal
1000 ✓	-8
1001 ✓	-7
1010 ✓	-6
1011	-5
1100	-4
1101	-2
1110	-2
1111	-1

2's Comp Repr

1001

2's Comp

0111 → -7

2's Comp Repr

1000 →

↓

1000 → -8

1010
↓
0110
6

-6

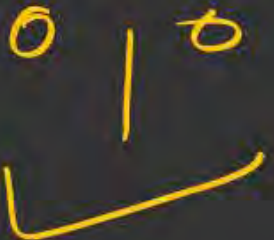
-6 ⇒ 0110
1010

#Q. If the number given below are written in 1's complement convert it into Equivalence Decimal?

- (1) 01101 $\rightarrow +13$ ✓
 (3) 00001101 $\rightarrow +13$ ✓
 (5) 11110010 $\rightarrow -13$ ✓
 (7) 0000 $\rightarrow 0$

- (2) 001101 $\rightarrow +13$ ✓
 (4) 110010 $\rightarrow -13$ ✓
 (6) 11101010 $\rightarrow -21$
 (8) 1111 $\rightarrow -0$

1111 1's
0000

Tricky Questions 

[MCQ]



#Q. F's complement of $(2BFD)_{\text{hex}}$ is:

88.9%

A E304 ~~X~~

B D403 ~~X~~

C D402

D C402 ~~X~~

$$\begin{array}{r} F F F F \\ - 2 B F D \\ \hline D 4 0 2 \\ \hline \end{array}$$

#Q. The greatest(negative number, which can be stored in a computer that has 8-bit word length and uses 2's complement arithmetic, is

237.

A -256 X

B -255 X

C -128 ✓

D -127 X

(C)

2's Comp n bit

$$-\{2^{n-1}\} \text{ to } (2^{n-1} - 1)$$

$$-2^{8-1} = -2^7 = -128$$

#Q. 11001, 1001 and 111001 correspond to the 2's complement representation of which one of the following sets of number?

1111001

-771

- A** 25, 9 and 57 respectively $\times$
- B** -6, -6 and -6 respectively $\times$
- C** -7, -7 and -7 respectively
- D** -25, -9 and -57 respectively $\times$

11001 $\rightarrow$ -7

$\downarrow$
00111

$\leftarrow$
1001 $\rightarrow$ -7

$\downarrow$
0111

$\leftarrow$
111001 $\rightarrow$ -7
 $\downarrow$
000111

[MCQ]

Tricky



#Q. P, Q and R are the decimal integers corresponding to the 4-bit binary number 1100 considered in signed magnitude, 1's complement and 2's complement representations, respectively.

The 6-bit 2's complement representation of $(P + Q + R)$ is

37%

A 111101 ✗

B 110101 ✓

C 110010 ✗

D 111001 ✗

70%
↓
37.5%



Soln:-

$$N = 1100$$

P $\rightarrow$ Signed Magnitude of N

Q $\rightarrow$ 1's Comp Repr of N

R $\rightarrow$ 2's Comp Repr of N

$$1100 \rightarrow \begin{array}{c} \text{---}4 \\ \downarrow \\ \boxed{0100} +4 \\ \text{R} = -4 \end{array}$$

$$\boxed{1100} \\ \text{P} = \underline{\underline{-4}}$$

$$1100 \rightarrow -3 \quad \boxed{Q = -3}$$
$$\begin{array}{c} 1100 \\ \downarrow \\ \boxed{0011} +3 \end{array}$$

[MCQ]



#Q. A number is expressed in binary two's complement as 10011. Its decimal equivalent value is:

95%
↓
90%

①0011
↓
01101 -13

A 19 ✗

B 13 ✗

C -19 ✗

☒ **D** -13

[MCQ]



#Q. What is the possible base of the given number system?

$$\sqrt{41} = 5$$

80%
54%

$$\sqrt{41} = 5$$

Ans - 6

- A** 4
- B** 5
- C** 10
- D** 6

[MCQ]



#Q. Find the base values which can satisfy the following equation.

$$\left(\frac{39}{3} = 13\right)_x$$

A

Only 10

B

Any base

C

Any base > 9

D

Any base ≥ 9

[MCQ]



#Q. Find the base values which can satisfy the following two equations simultaneously.

(i) $2 + 3 = 5$

(ii) $2 \times 4 = 10$

A

Any base ≥ 9

B

8

C

Any base ≥ 8

D

6

[MCQ]



#Q. Find number of solutions of x and y which can satisfy the equation $(43)_8 = (x8)_y$

A

1

~~**B**~~

2

C

3

D

4

$x \quad y$
 $\left\{ \begin{array}{ll} 1 & 27 \checkmark \\ 3 & 9 \checkmark \end{array} \right.$
 $\left(\begin{array}{ll} 9 & 3 \\ 27 & 1 \end{array} \right) \times$

$$(43)_8 = (x8)_y$$
$$\downarrow \quad \downarrow$$
$$4 \times 8^1 + 3 \times 8^0 = x \times y + 8$$

$$32 + 3 = xy + 8$$

$$xy = 27$$

90%

$$\underline{y > x \text{ \& } y > 8}$$

[MCQ]



V. Imp

#Q. Find number of solutions of x and y which can satisfy the equation
 $(123)_5 = (x8)_y$



(y > x and y > 8 for valid)

50%

A 1 X

B 2 X

C 3

D 4 X

Soln: $(123)_5 = (x8)_y$

$\downarrow$ $\downarrow$
 $()_{10}$ $()_{10}$

$$1 \times 5^2 + 2 \times 5^1 + 3 \times 5^0 = x \times y^1 + 8 \times y^0$$

$$25 + 10 + 3 = xy + 8$$

$$38 = xy + 8$$

$$xy = 30$$

Imp

$(x8)_y$

$x < y$
 $8 < y$

	x	y		
{	1	30	✓	✓
	2	15	✓	✓
	3	10	✓	✓

5	6
6	5
10	3
15	2
30	1

X

[MCQ]



#Q. The result of addition operation $34 + 43$ performed on minimum base is stored in an 8-bit register. The content of register will be

A

01000011

B

00101010

C

01010101

D

01010100



2 mins Summary



Number System ✓



THANK - YOU